

## PRILEP, North Macedonia

WMO: 135850

Lat: 41.333N

Lon: 21.567E

Elev: 674

StdP: 93.49

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-----|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |     |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p) |
| 1             | -11.4      | -8.1 | -14.3                         | 1.2 | -9.5 | -11.7 | 1.5 | -5.8 | 14.0                  | 7.4  | 11.2 | 3.8  | 0.7                      | 0    | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.3                   | 33.7            | 19.1 | 31.9 | 18.7 | 30.3 | 18.2 | 20.3                | 29.9 | 19.6 | 28.9 | 19.0 | 27.9 | 2.3                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.7                            | 13.8 | 21.7 | 16.9 | 13.1 | 21.5 | 16.1 | 12.4 | 21.3 | 61.3          | 30.1 | 58.9 | 28.8 | 56.6 | 28.0 | 22.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.4          | 8.5  | 7.0 | -14.0                      | 36.4 | 3.5                | 2.3 | -16.5                                              | 38.0 | -18.5      | 39.3 | -20.4      | 40.6 | -22.9      | 42.3 |    |    |
| (5)               |      |     | -14.4                      | 21.5 | 3.5                | 0.8 | -16.9                                              | 22.1 | -19.0      | 22.5 | -21.0      | 23.0 | -23.6      | 23.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.3        | 0.6   | 3.4   | 7.6   | 11.8  | 15.9  | 20.1  | 23.0  | 23.1  | 18.4  | 13.1  | 8.2   | 2.2   |  |  |
|                                                                        | DBStd       | 8.52        | 4.87  | 4.69  | 3.75  | 3.63  | 3.09  | 3.14  | 2.92  | 2.59  | 3.23  | 3.76  | 3.90  | 4.89  |  |  |
|                                                                        | HDD10.0     | 929         | 291   | 187   | 93    | 22    | 1     | 0     | 0     | 0     | 0     | 17    | 77    | 243   |  |  |
|                                                                        | HDD18.3     | 2606        | 549   | 417   | 333   | 198   | 86    | 16    | 3     | 1     | 39    | 163   | 304   | 498   |  |  |
|                                                                        | CDD10.0     | 1786        | 1     | 2     | 18    | 74    | 184   | 304   | 403   | 406   | 251   | 114   | 23    | 3     |  |  |
|                                                                        | CDD18.3     | 421         | 0     | 0     | 0     | 1     | 12    | 70    | 148   | 149   | 40    | 2     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 4127        | 0     | 0     | 0     | 16    | 129   | 681   | 1513  | 1422  | 337   | 31    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 1542        | 0     | 0     | 0     | 1     | 23    | 224   | 629   | 574   | 88    | 3     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.8         | 1.6   | 1.9   | 2.7   | 2.2   | 1.8   | 1.9   | 2.0   | 1.8   | 1.7   | 1.5   | 1.5   | 1.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 578         | 44    | 42    | 45    | 49    | 70    | 52    | 33    | 33    | 42    | 52    | 65    | 50    |  |  |
|                                                                        | PrecMax     | 752         | 107   | 124   | 106   | 108   | 123   | 127   | 90    | 116   | 104   | 135   | 226   | 104   |  |  |
|                                                                        | PrecMin     | 414         | 15    | 4     | 7     | 23    | 8     | 11    | 0     | 0     | 3     | 0     | 13    | 4     |  |  |
|                                                                        | PrecStd     | 96          | 24    | 31    | 25    | 23    | 33    | 36    | 30    | 26    | 31    | 36    | 53    | 28    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 13.5  | 17.9  | 20.8  | 25.5  | 29.4  | 34.1  | 36.3  | 35.8  | 31.5  | 26.5  | 20.5  | 16.4  |  |  |
|                                                                        |             | MCWB        | 7.8   | 9.4   | 10.6  | 13.9  | 16.8  | 19.1  | 19.7  | 19.6  | 17.9  | 15.9  | 12.9  | 10.8  |  |  |
|                                                                        | 2%          | DB          | 10.9  | 14.1  | 18.6  | 22.5  | 26.6  | 31.3  | 34.0  | 33.7  | 29.3  | 23.7  | 18.0  | 12.8  |  |  |
|                                                                        |             | MCWB        | 6.5   | 7.9   | 9.6   | 12.9  | 15.9  | 18.6  | 19.1  | 19.2  | 17.1  | 14.4  | 12.0  | 9.1   |  |  |
|                                                                        | 5%          | DB          | 9.0   | 12.1  | 16.6  | 20.4  | 24.7  | 29.4  | 32.0  | 31.8  | 27.1  | 21.9  | 16.0  | 10.6  |  |  |
|                                                                        |             | MCWB        | 5.7   | 7.1   | 8.9   | 11.9  | 15.1  | 17.9  | 18.7  | 18.7  | 16.3  | 13.6  | 10.6  | 7.7   |  |  |
|                                                                        | 10%         | DB          | 7.2   | 10.1  | 14.5  | 18.4  | 22.8  | 27.5  | 30.3  | 30.0  | 25.2  | 19.7  | 14.1  | 8.7   |  |  |
|                                                                        |             | MCWB        | 4.8   | 6.0   | 7.8   | 10.9  | 14.4  | 17.3  | 18.3  | 18.2  | 15.6  | 13.1  | 10.1  | 6.7   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 8.7   | 10.4  | 11.5  | 15.0  | 18.3  | 20.5  | 21.1  | 21.6  | 19.2  | 16.6  | 14.2  | 11.2  |  |  |
|                                                                        |             | MCDB        | 11.7  | 15.4  | 17.7  | 22.9  | 26.1  | 29.5  | 31.0  | 31.5  | 27.8  | 23.5  | 18.1  | 14.9  |  |  |
|                                                                        | 2%          | WB          | 7.1   | 8.9   | 10.4  | 13.8  | 16.8  | 19.5  | 20.3  | 20.4  | 18.0  | 15.6  | 12.6  | 9.6   |  |  |
|                                                                        |             | MCDB        | 9.7   | 13.2  | 17.0  | 20.3  | 24.0  | 28.6  | 29.8  | 30.2  | 25.8  | 21.1  | 15.9  | 11.7  |  |  |
|                                                                        | 5%          | WB          | 6.1   | 7.6   | 9.3   | 12.8  | 16.0  | 18.7  | 19.7  | 19.6  | 17.2  | 14.7  | 11.7  | 8.4   |  |  |
|                                                                        |             | MCDB        | 8.7   | 11.2  | 15.3  | 18.6  | 22.5  | 27.3  | 28.9  | 29.0  | 24.4  | 19.3  | 14.9  | 10.2  |  |  |
|                                                                        | 10%         | WB          | 4.9   | 6.5   | 8.3   | 11.8  | 15.1  | 17.9  | 19.0  | 18.8  | 16.5  | 13.9  | 10.6  | 6.9   |  |  |
|                                                                        |             | MCDB        | 6.9   | 9.6   | 13.6  | 16.8  | 21.2  | 25.6  | 28.0  | 27.7  | 22.8  | 18.2  | 13.4  | 8.7   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.7   | 8.2   | 9.1   | 9.7   | 10.3  | 11.5  | 12.3  | 12.5  | 10.8  | 9.9   | 8.2   | 6.4   |  |  |
|                                                                        |             | MCDBR       | 9.5   | 12.0  | 13.6  | 13.3  | 13.8  | 14.5  | 15.5  | 15.0  | 14.2  | 13.7  | 11.5  | 8.7   |  |  |
|                                                                        | 5% WB       | MCWBR       | 5.9   | 6.4   | 6.9   | 5.9   | 5.4   | 5.2   | 5.6   | 5.4   | 5.0   | 5.8   | 6.1   | 5.5   |  |  |
|                                                                        |             | MCWBR       | 8.4   | 10.6  | 12.0  | 11.4  | 12.1  | 13.4  | 13.3  | 13.5  | 11.8  | 10.7  | 9.5   | 7.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.320 | 0.347 | 0.392 | 0.448 | 0.441 | 0.420 | 0.421 | 0.437 | 0.421 | 0.382 | 0.349 | 0.323 |  |  |
|                                                                        |             | taud        | 2.375 | 2.292 | 2.199 | 2.050 | 2.137 | 2.243 | 2.236 | 2.188 | 2.209 | 2.305 | 2.360 | 2.366 |  |  |
|                                                                        | Ebn at Noon |             | 824   | 852   | 849   | 824   | 839   | 857   | 851   | 825   | 810   | 800   | 778   | 785   |  |  |
|                                                                        |             | Edh at Noon | 83    | 105   | 130   | 161   | 152   | 137   | 136   | 139   | 127   | 103   | 84    | 78    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.71  | 2.58  | 3.61  | 4.67  | 5.66  | 6.52  | 6.83  | 6.09  | 4.44  | 3.11  | 2.01  | 1.48  |  |  |
|                                                                        | RadStd      |             | 0.25  | 0.32  | 0.44  | 0.54  | 0.36  | 0.48  | 0.39  | 0.38  | 0.37  | 0.35  | 0.21  | 0.26  |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (22 neighbors) | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -125    | N/A     | N/A     |

Nomenclature: See separate page